



Project 4

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In this project, we designed and implemented a comparator circuit that receives two 3-bit inputs (A2 A1 A0 and B2 B1 B0) and gives three output cases:

Boolean equations (cases):

1 $A=B$

1- $A_2=B_2$ AND $A_1=B_1$ AND $A_0=B_0$

2- $X_i = (A_i' . B_i' + A_i . B_i) = A_i(.)B_i$

$X_0 = A_0(.)B_0$, $X_1 = A_1(.)B_1$, $X_2 = A_2(.)B_2$

$X = X_2 . X_1 . X_0$

2 $A > B$

1- $(A_2=1$ AND $B_2=0)$

2- OR $(A_2=B_2)$ AND $(A_1=1$ AND $B_1=0)$

3- OR $(A_2=B_2$ AND $A_1=B_1)$ AND $(A_0=1$ AND $B_0=0)$

$Y = A_2 . B_0' + X_2 . A_1 . B_1' + X_2 . X_1 . A_0 . B_0'$

3 **A < B**

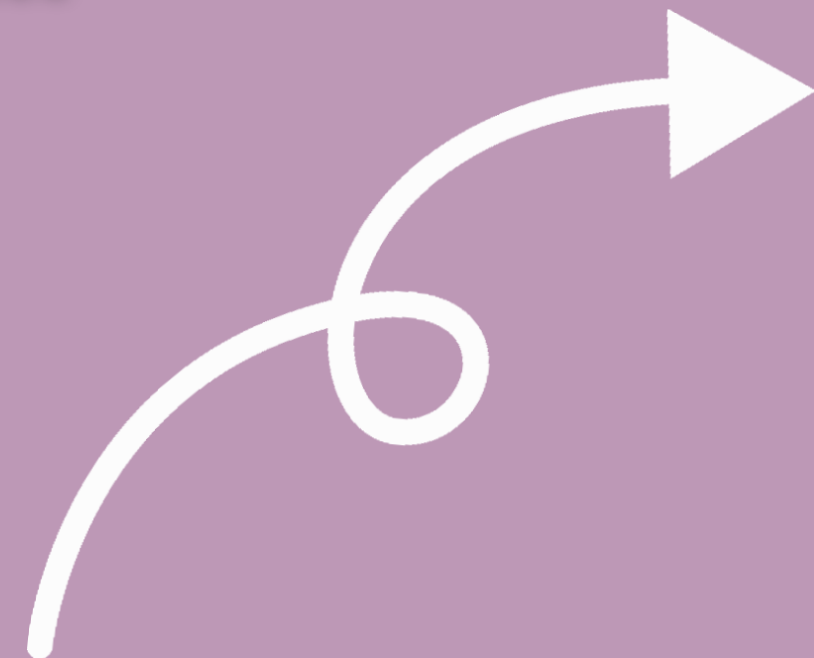
1- $(A_2=0 \text{ AND } B_2=1)$

2- $\text{OR } (A_2=B_2) \text{ AND } (A_1=0 \text{ AND } B_1=1)$

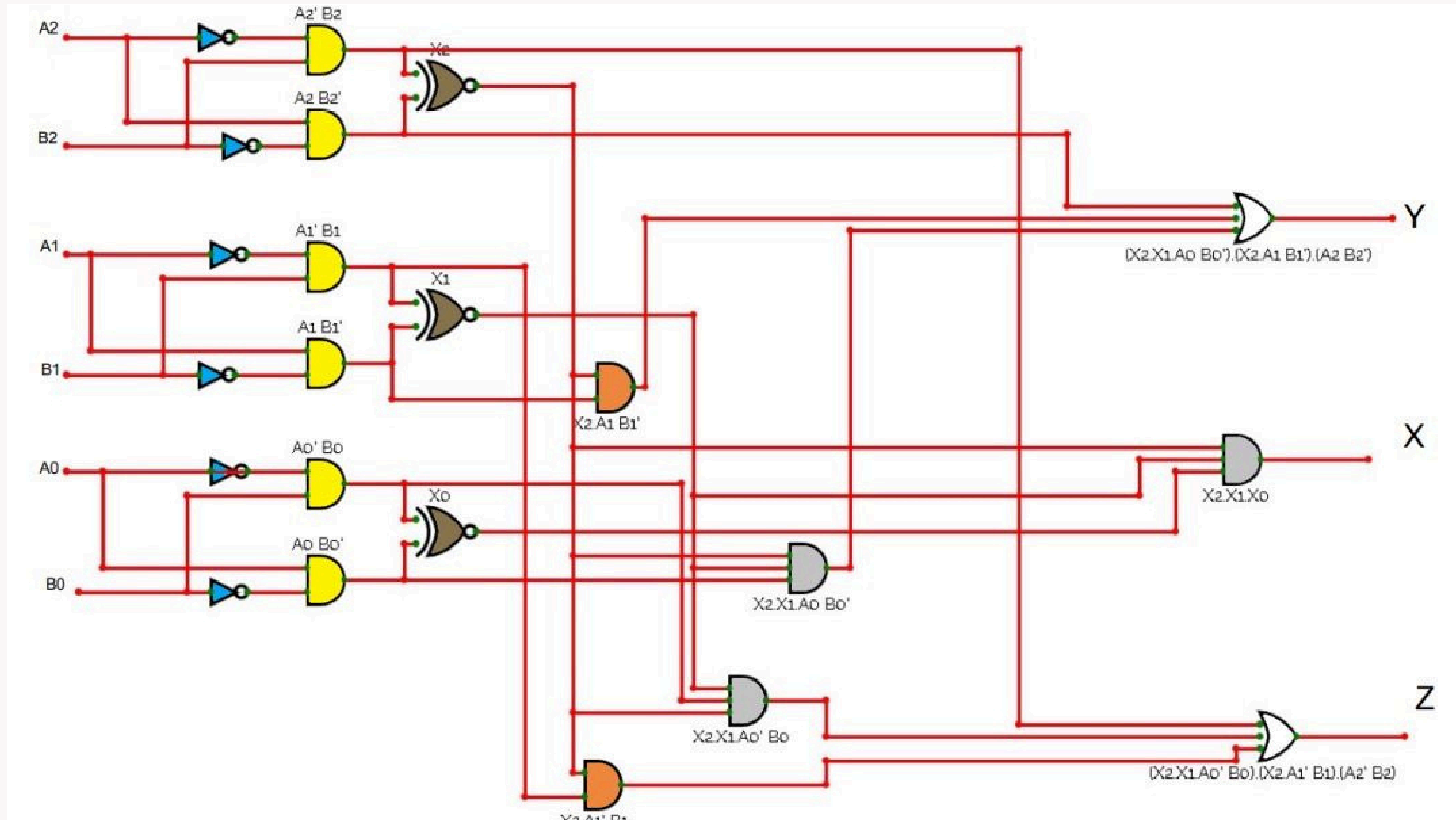
3- $\text{OR } (A_2=B_2 \text{ AND } A_1=B_1) \text{ AND } (A_0=0 \text{ AND } B_0=1)$

$$Z = (A_2' \cdot B_2) + (X_2 \cdot A_1' \cdot B_1) + (X_1 \cdot X_2 \cdot A_0' \cdot B_0)$$

Then we visualized our equations into a simple diagram



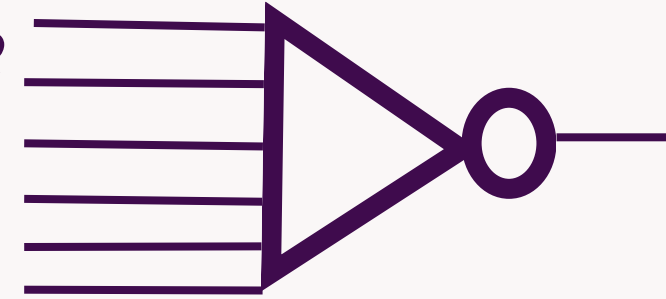
Logic Diagram:



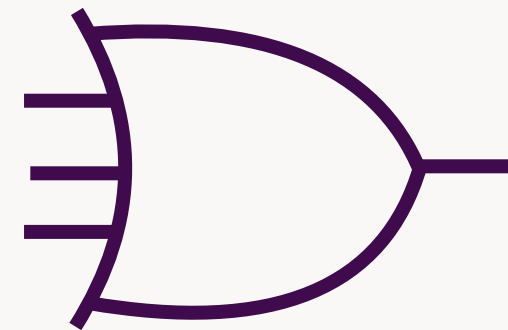
Components used:

**After carefully solving the Boolean equations,
we deduced that the system requires:**

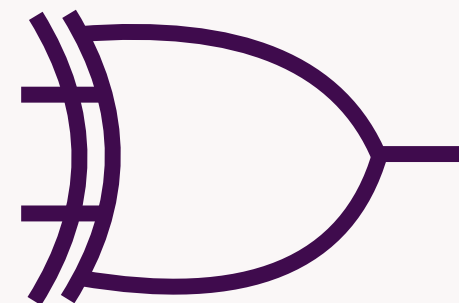
- **1) One Not Gate**
with 6 inputs



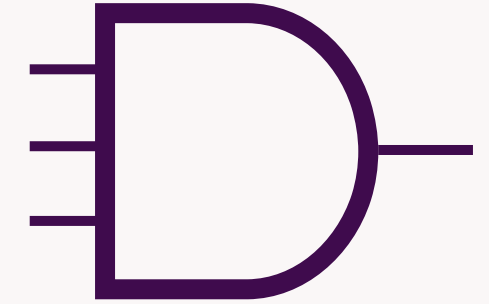
- **2) One OR Gate**
with 3 inputs



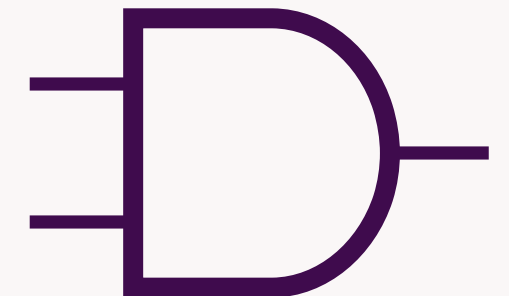
- **3) One Xnor Gate**
with 2 inputs



- **4) One AND Gate**
with 3 inputs



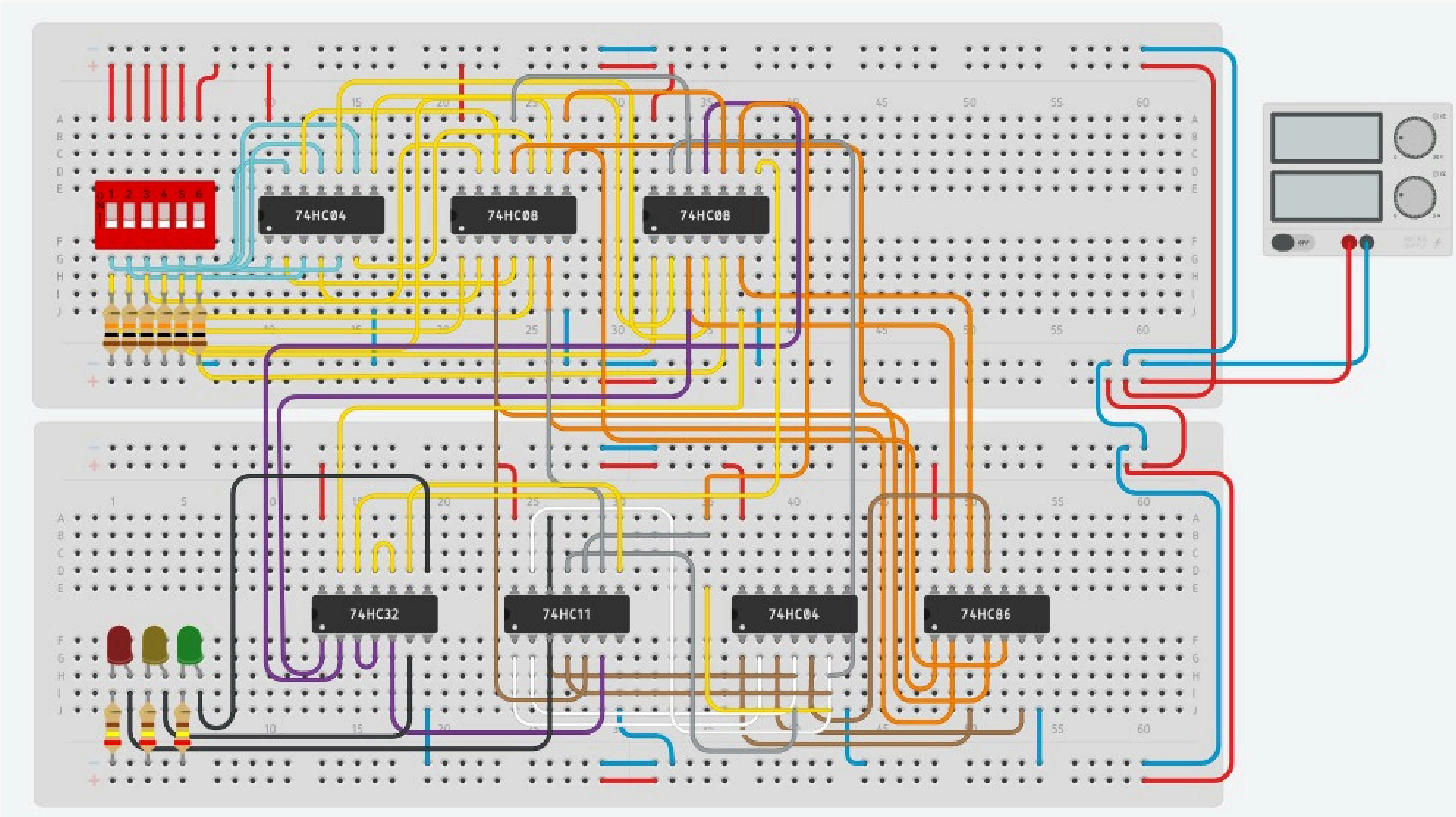
- **5) One AND Gate**
with 2 inputs



Other components used:

Item	usage
Dip switch	<i>Turns the circuit on and off</i>
Battery 9v/Power supply	<i>Provides the circuit with electricity</i>
Resistors 10K / 240 ohm	<i>Decreases the voltage to 5V</i>
2 Bread Boards	<i>Used to implement and design the circuit</i>

Circuit simulation:



THANK
YOU!